

MAMATHA BANDARU KISHANKUMAR

Data Engineer

Phone: +1 (682) 738 6642 | E-mail: m.bandaruk.bk@gmail | [LinkedIn](#) | Location: Irving, Texas

SUMMARY

- Data Engineer with 4+ years of experience in designing, developing, and optimizing data pipelines for large-scale data systems.
- Expertise in Big Data Technologies including Apache Hadoop, Spark, Kafka, and cloud platforms like AWS, Google Cloud, and Microsoft Azure.
- Skilled in building scalable ETL pipelines using tools like Apache Spark, AWS Glue, and Talend for seamless data integration and transformation.
- Proficient in Data Warehousing solutions including Snowflake, Redshift, and BigQuery for efficient data storage and retrieval.
- Experienced in real-time data streaming and processing using Apache Kafka and AWS Lambda, enhancing data flow and enabling timely analytics.
- Advanced knowledge of SQL, Python, PySpark, and R, with the ability to create complex queries and data transformations.
- Strong background in data governance frameworks such as GDPR and HIPAA, ensuring compliance and secure data management.
- Proven ability to deliver actionable insights through data visualization using tools like Power BI, Tableau, and Jupyter Notebook.
- Adept in DevOps tools such as Docker, Kubernetes, Jenkins, and Terraform, optimizing the deployment and management of data workflows.

SKILLS

- **Programming Languages:** Java, Python, PySpark, Scala, R, SQL, Shell Scripting
- **Big Data Technologies:** Apache Hadoop, Apache Spark, Kafka, Databricks
- **Databases:** MySQL, PostgreSQL, SQL Server, MongoDB, Oracle, Elasticsearch, PL/SQL
- **Data Integration & ETL Tools:** Talend, Informatica PowerCenter, Databricks, AWS Glue
- **Cloud Platforms & Services:** AWS, Google Cloud Platform, Microsoft Azure, RDS
- **Data Warehousing:** Snowflake, Redshift, BigQuery
- **DevOps & Infrastructure:** Docker, Kubernetes, Jenkins, GitHub Actions, Terraform, Ansible
- **Data Modeling:** OLAP, OLTP, Data Lakes
- **Data Analysis & Visualization:** Power BI, Jupyter Notebook, Dashboard Creation, Data Collection, Data Analysis, Quality Control
- **Version Control:** Git, Bitbucket
- **Project & Process Management:** Jira, Trello, Asana
- **Data Governance Frameworks:** GDPR, HIPAA, EMR
- **Certifications:** AWS Solution Architect Associate, Udacity Frontend Nano Degree

WORK EXPERIENCE

Dell Technologies

Aug 2024 – Present

Data Engineer

- Spearheaded the development and deployment of a next-generation data Lakehouse solution, integrating the benefits of both data lakes, data warehouses, and Snowflake to streamline data storage and retrieval.
- Leveraged Apache Spark, AWS Glue, and Delta Lake to build high-performance ETL pipelines, reducing data processing latency by 30%.
- Worked closely with various teams to integrate real-time data streams via Apache Kafka, ensuring continuous and efficient data flow across platforms.
- Automated the data pipeline process and implemented rigorous data quality protocols, enabling reliable analytics and enhanced reporting capabilities.
- Established secure data access protocols and governance frameworks using AWS IAM, ensuring compliance with regulatory standards like GDPR and HIPAA.

CVS Health

Jan 2024 – May 2024

Data Engineer – Intern

- Engineered a robust data pipeline using Apache Spark and Python to integrate patient data from electronic health records (EHR) and other healthcare systems (EMR), enhancing data accuracy and reducing integration time by 25%.
- Utilized SQL and PostgreSQL to design complex queries for patient data analysis, enabling seamless extraction and transformation of large datasets for personalized healthcare insights.
- Utilized AWS S3 and Lambda to streamline data ingestion workflows, enhancing real-time processing efficiency and enabling prompt healthcare decision-making.

Infosys BPM	Jan 2021 – Dec 2022
Data Engineer	
- Developed and fine-tuned ETL workflows using Apache Spark and Python to handle and analyse retail customer behaviour data from diverse sources, enhancing processing efficiency by 20%. - Integrated and stored customer data in AWS Redshift and RDS, facilitating faster retrieval and in-depth analysis of customer trends. - Crafted advanced SQL queries to extract, filter, and consolidate customer interaction data, providing actionable insights for marketing and product teams. - Created interactive Power BI dashboards to visualize key behaviour metrics, empowering data-driven decisions across the retail business. - Leveraged Apache Kafka for real-time data ingestion, ensuring the latest customer insights were always available. - Streamlined data processing with Airflow, automating workflows and ensuring efficient, timely delivery of data. - Performed data cleansing and transformation using Python to guarantee the accuracy and consistency of datasets for customer behaviour analysis. - Worked closely with data scientists to implement machine learning models that predicted customer purchasing patterns and enhanced retail strategies.	

Trigent	Aug 2019 – Dec 2020
Data Engineer	
- Directed the shift of an enterprise data warehouse from conventional SQL systems to Hadoop, resulting in a 30% reduction in storage costs and increased efficiency. - Engineered and refined ETL pipelines using Apache Hive, Pig, Spark, and Databricks to ensure seamless data integration within the Hadoop ecosystem. - Implemented HBase for fast, low-latency data storage, improving the speed of data access and enabling efficient analysis. - Streamlined data flow automation with Apache NiFi, minimizing manual processes and increasing operational speed. - Worked alongside business and technical teams to ensure the migration process met both business needs and regulatory standards. - Performed comprehensive data accuracy checks and validation to ensure consistency during migration. - Optimized the performance of Hadoop clusters, achieving a 25% reduction in query execution time and boosting overall system performance.	

PROJECTS

Real-Time Retail Sales Monitoring	
- Developed a Kafka-Snowflake pipeline, reducing data processing latency by 20% and enabling real-time sales monitoring for 100+ retail outlets. - Designed dynamic dashboards in Power BI to visualize inventory levels and predict stock shortages.	
Healthcare Claims Analytics	
- Developed a scalable ETL pipeline using AWS Glue to process and transform claims data for a healthcare provider. - Integrated Elasticsearch for anomaly detection in claims, enabling early identification of fraudulent activities.	
IoT Sensor Data Integration	
- Optimized Apache Spark and MongoDB pipeline, reducing data processing time by 40% for IoT sensor analytics. - Visualized key metrics in Tableau, enabling predictive maintenance for industrial equipment.	

EDUCATION

Master in Computer Science	Jan 2023 – May 2024
California State University Long Beach	
Bachelor of Technology in Computer Science	Jun 2017 – May 2021
Madanapalle Institute of Technology and Science	